Turbulent Heat Fluxes from a Prescribed Friction Velocity

A Non-Circular Bulk Algorithm with Wind-Dependent Charnock Roughness

Algorithm description and user guide

August 16, 2026

Abstract

This document describes a diagnostic bulk flux algorithm that computes the sensible heat flux H_S and latent heat flux H_L from a *prescribed* friction velocity u_* together with bulk meteorological state (sea surface temperature, air temperature, relative humidity, pressure). The intended application is satellite-retrieved u_*, for example from GNSS-reflectometry, where the friction velocity is an observable and the near-surface wind speed is not.

The algorithm follows COARE-type Monin–Obukhov similarity for the scalar fluxes, but differs from a standard bulk algorithm in one essential respect: because u_* is an input rather than an output, any roughness parameterisation that depends on wind speed risks circularity, since the only available wind speed would be one reconstructed from u_* and z_0 themselves. The algorithm avoids this by inferring a proxy neutral wind $\widehat{U}_{10N}$ through the inversion of an *independent* prior relation $u_*(U_{10N})$, and applying the Charnock coefficient law to that proxy. No internally computed wind ever feeds back into the roughness.

Sections 3–6 give the governing equations, the five available roughness closures, and the numerical solution. Section 7 specifies the validity and quality-control semantics, which are deliberately strict: a solve that does not converge does not produce a flux. Section 8 gives a quantitative uncertainty budget, whose central result is that with u_* prescribed the momentum roughness reaches the flux only through a logarithm, and over an identifiable low-wind regime does not reach it at all.

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1 Scope and design intent

1.1 What the algorithm does

Given, for each observation, a friction velocity u_* and the bulk state $(T_s, T_a, \mathrm{RH}, P)$, the algorithm returns the turbulent heat fluxes

$$H_S = -\rho_a c_p u_* \theta_*, \qquad H_L = -\rho_a L_e u_* q_*, \tag{1}$$

with the sign convention that positive values denote an *upward* flux (ocean → atmosphere). The temperature and humidity scaling parameters θ_* and q_* are obtained from Monin–Obukhov similarity theory, solved iteratively as described in Section 6.

1.2 What is an input, and what is only a diagnostic

The defining constraint is that u_* is **fixed**. It is never adjusted by the iteration. This has three consequences that a user must keep in mind:

1. The momentum flux $\tau = \rho_a u_*^2$ is a restatement of the input, not an independent result.
2. The mean wind $\overline{U}(z_u)$ and neutral wind U_{10N} reported in the output are *diagnostics* reconstructed from u_* and z_0. They must never be regressed against, or validated against, quantities that were themselves used to construct them.

3. Gustiness cannot influence the heat fluxes. In a standard bulk algorithm the gustiness velocity U_g enters through the effective wind that determines u_*; here u_* is given, so U_g has no pathway into Eq. (1). The gustiness variables are computed and reported for diagnostic continuity only. This is verified numerically: enabling and disabling gustiness produces bit-identical H_S and H_L.

1.3 Deliberate omissions

The following are *not* included, and results should not be described as COARE-equivalent on their account: the Webb correction to H_L; the cool-skin and warm-layer parameterisations; and the sensible heat flux carried by rainfall. Each is a physically meaningful term that a full COARE implementation would carry.

2 Notation

Symbol	Meaning	Units
u_*	friction velocity (prescribed input)	$\mathrm{m\,s^{-1}}$
$\theta_*,\ q_*$	temperature / humidity scaling parameters	K, $\mathrm{kg\,kg^{-1}}$
$T_s,\ T_a$	sea surface and air temperature	°C
T_{aK}	air temperature, absolute	K
RH	relative humidity	%
P	atmospheric pressure	hPa or Pa
S	sea surface salinity	psu
$Q,\ Q_s$	specific humidity of air, saturation at surface	$\mathrm{kg\,kg^{-1}}$
$\Delta\theta$	air–sea potential temperature difference	K
Δq	air–sea humidity difference	$\mathrm{kg\,kg^{-1}}$
z_0	momentum roughness length	m
$z_{0t},\ z_{0q}$	scalar roughness lengths (heat, moisture)	m
z_u, z_t, z_q	measurement heights (wind, temp., humidity)	m
ζ	stability parameter z_u/L	–
L	Obukhov length	m
$\psi_m,\ \psi_h$	integrated stability functions	–
Rr	roughness Reynolds number $z_0 u_*/\nu$	–
α	Charnock coefficient	–
β_v	viscous roughness coefficient (= 0.11)	–
ν	kinematic viscosity of air	$\mathrm{m^2\,s^{-1}}$
ρ_a	moist air density	$\mathrm{kg\,m^{-3}}$
c_p	specific heat of dry air (= 1004.67)	$\mathrm{J\,kg^{-1}\,K^{-1}}$
L_e	latent heat of vaporisation	$\mathrm{J\,kg^{-1}}$
κ	von Kármán constant (= 0.4)	–
g	gravity at latitude	$\mathrm{m\,s^{-2}}$
H_s	significant wave height	m
$T_p,\ L_p$	peak wave period and wavelength	s, m
c_p^w	phase speed at the spectral peak	$\mathrm{m\,s^{-1}}$
U_{10N}	10 m neutral wind speed	$\mathrm{m\,s^{-1}}$
$\widehat{U}_{10N}$	*proxy* U_{10N} from prior inversion	$\mathrm{m\,s^{-1}}$

3 Governing equations

3.1 Flux–profile relations

With u_* prescribed, the scalar scaling parameters follow directly from the integrated flux–profile relations:

$$\theta_* = \frac{-\kappa\,\Delta\theta}{\ln(z_t/z_{0t}) - \psi_h(\zeta\, z_t/z_u)}, \tag{2}$$

$$q_* = \frac{-\kappa\,\Delta q}{\ln(z_q/z_{0q}) - \psi_h(\zeta\, z_q/z_u)}. \tag{3}$$

The gradients use a COARE-style adiabatic lapse correction applied at the temperature measurement height,

$$\Delta\theta = T_s - T_a - \Gamma z_t, \qquad \Gamma = 0.0098\ \mathrm{K\,m^{-1}}, \qquad \Delta q = Q_s - Q. \tag{4}$$

3.2 Stability

The stability parameter is defined at the wind reference height z_u,

$$\zeta \;\equiv\; \frac{z_u}{L} \;=\; \frac{\kappa\, g\, z_u}{T_{aK}} \cdot \frac{\theta_* + 0.61\, T_{aK}\, q_*}{u_*^2}, \qquad L = \frac{z_u}{\zeta}. \tag{5}$$

Two properties of Eq. (5) matter for what follows. First, ζ contains **no roughness length**: with u_* prescribed, stability is determined entirely by the scaling parameters and the input friction velocity. Second, $\zeta \propto u_*^{-2}$, so the stability parameter diverges as $u_* \to 0$. Section 6.5 treats the consequences.

Because ζ is referenced to z_u, the stability functions in Eqs. (2)–(3) are evaluated at the rescaled arguments $\zeta z_t/z_u$ and $\zeta z_q/z_u$. When the three measurement heights are equal these reduce to $\psi_h(\zeta)$.

3.3 Stability functions

The algorithm uses the COARE 3.0 forms. For the scalar function ψ_h, in stable conditions ($\zeta \geq 0$), with $d = \min(50,\, 0.35\zeta)$:

$$\psi_h = -\left[\left(1 + \tfrac{2}{3}\zeta\right)^{3/2} + \tfrac{2}{3}\,(\zeta - 14.28)\, e^{-d} + 8.525\right]. \tag{6}$$

In unstable conditions ($\zeta < 0$) a Kansas-type form ψ_k and a convective form ψ_c are blended:

$$x_k = (1 - 15\zeta)^{1/2}, \qquad \psi_k = 2\ln\!\left(\frac{1 + x_k}{2}\right), \tag{7}$$

$$x_c = (1 - 34.15\zeta)^{1/3}, \qquad \psi_c = \tfrac{3}{2}\ln\!\left(\frac{1 + x_c + x_c^2}{3}\right) - \sqrt{3}\arctan\!\left(\frac{1 + 2x_c}{\sqrt{3}}\right) + \frac{\pi}{\sqrt{3}}, \tag{8}$$

$$\psi_h = (1 - f)\,\psi_k + f\,\psi_c, \qquad f = \frac{\zeta^2}{1 + \zeta^2}. \tag{9}$$

The momentum function ψ_m, used only for the diagnostic wind, takes the analogous form with $a = 0.7$, $b = 3/4$, $c = 5$, $d = 0.35$:

$$\psi_m\big|_{\zeta \geq 0} = -\left[a\zeta + b\,(\zeta - \tfrac{c}{d})\, e^{-\min(50,\, 0.35\zeta)} + \tfrac{bc}{d}\right], \tag{10}$$

and for $\zeta < 0$ the same blend with $x_k = (1 - 15\zeta)^{1/4}$, $x_c = (1 - 10.15\zeta)^{1/3}$ and $\psi_k = 2\ln(\frac{1+x_k}{2}) + \ln(\frac{1+x_k^2}{2}) - 2\arctan(x_k) + \frac{\pi}{2}$.

4 Thermodynamic preliminaries

Saturation vapour pressure uses the Buck formulation, with a separate branch below the freezing point T_f (e in hPa, T in °C, P in hPa):

$$e_{\mathrm{sat}}(T,P) = \begin{cases} 6.1121\,\exp\!\left(\dfrac{17.502\,T}{T+240.97}\right)\left(1.0007+3.46\times10^{-6}P\right), & T \ge T_f, \\ 6.1115\,\exp\!\left(\dfrac{22.452\,T}{T+272.55}\right)\left(1.0003+4.18\times10^{-6}P\right), & T < T_f. \end{cases} \tag{11}$$

For air, $T_f = 0$°C. For sea water the freezing point is depressed by salinity, and the surface saturation vapour pressure is reduced by the salinity factor f_s:

$$T_f(S) = -0.0575\,S + 1.71052\times10^{-3}S^{3/2} - 2.154996\times10^{-4}S^2, \qquad f_s = 1 - 0.02\,\frac{S}{35}. \tag{12}$$

Specific humidities follow from the vapour pressures:

$$Q = \frac{0.622\,e_a}{P - 0.378\,e_a}, \quad e_a = (\mathrm{RH}/100)\,e_{\mathrm{sat}}(T_a,P); \qquad Q_s = \frac{0.622\,f_s\,e_{\mathrm{sat}}(T_s,P)}{P - 0.378\,f_s\,e_{\mathrm{sat}}(T_s,P)}. \tag{13}$$

The remaining thermodynamic quantities are

$$\rho_a = \frac{P_{\mathrm{Pa}}}{R_d\,T_{aK}\,(1+0.61Q)}, \qquad R_d = 287.05\ \mathrm{J\,kg^{-1}\,K^{-1}}, \tag{14}$$

$$L_e = (2.501 - 0.00237\,T_s)\times10^6, \tag{15}$$

$$\nu = 1.326\times10^{-5}\Big(1 + 6.542\times10^{-3}\,T_a + 8.301\times10^{-6}\,T_a^2 - 4.84\times10^{-9}\,T_a^3\Big). \tag{16}$$

Gravity is evaluated at the observation latitude using the normal-gravity series $g(\phi) = \gamma\,(1 + c_1x^2 + c_2x^4 + c_3x^6 + c_4x^8)$ with $x = \sin\phi$, $\gamma = 9.7803267715$.

Specific heat. The sensible heat flux is built from the **dry-air** value $c_p = 1004.67\ \mathrm{J\,kg^{-1}\,K^{-1}}$, consistent with COARE 3.5, in which the moist value $c_{pv} = c_p(1 + 0.84\,Q)$ is a distinct quantity not used in H_S. An option exists to substitute c_{pv}; it shifts H_S by approximately +0.8% at $Q \approx 0.01\,\mathrm{kg\,kg^{-1}}$. Note that the coefficient is 0.84 for Q in $\mathrm{kg\,kg^{-1}}$; the frequently quoted 0.00084 presumes Q in $\mathrm{g\,kg^{-1}}$.

5 Surface roughness

5.1 The circularity problem

A wind-dependent Charnock coefficient $\alpha(U_{10N})$ is standard in modern bulk algorithms. Applying it here is not straightforward. The algorithm has no observed wind speed: the only wind available is the diagnostic

$$U_{10N}^{\mathrm{diag}} = \frac{u_*}{\kappa}\ln\!\left(\frac{10}{z_0}\right), \tag{17}$$

which is itself a function of z_0. Using Eq. (17) to select α, and hence z_0, would close a loop in which the roughness is tuned by a wind that the roughness produced. Any subsequent regression of fluxes against U_{10N}^{diag} would then partly recover the assumed parameterisation rather than a physical relationship.

5.2 Resolution: inversion of an independent prior

The algorithm breaks the loop by introducing an *independent*, externally specified relation $\mathcal{F}: U_{10N} \mapsto u_*$, calibrated elsewhere and never informed by the present computation. Given the observed u_*, a proxy neutral wind is obtained by inverting it:

$$\boxed{\widehat{U}_{10N} = \mathcal{F}^{-1}(u_*), \qquad \alpha = \alpha\Big(\widehat{U}_{10N}\Big)\,.} \tag{18}$$

The proxy $\widehat{U}_{10N}$ is a monotone relabelling of the observed u_* and carries no information beyond it. Crucially it does *not* depend on z_0, so the roughness closure is a function of the input alone. Two priors are provided.

Andreas-type prior.

$$\mathcal{F}(U) = 0.239 + 0.0433\left[(U - 8.271) + \sqrt{0.120\,(U - 8.271)^2 + 0.181}\right]. \tag{19}$$

This relation is strictly increasing but has a non-zero floor: $\mathcal{F}(0) = 0.006287\,\mathrm{m\,s^{-1}}$. Over $U_{10N} \in [0, 60]$ its range is $u_* \in [0.00629,\ 3.255]$.

Edson-type prior. A logistic blend between a low-wind and a fully-rough branch,

$$\mathcal{F}(U) = (1 - w)\,s_\ell U + w\,(s_h U + b_h)\,, \qquad w = \left[1 + \exp\!\Big(-\frac{U - U_2}{\Delta}\Big)\right]^{-1}, \tag{20}$$

with defaults $s_\ell = 0.035$, $s_h = 0.062$, $b_h = -0.28$, $U_2 = 8.5$, $\Delta = 0.6$. This blend is strictly increasing on $[0, 60]$.

Numerical inversion. Because $\mathcal{F}$ is monotone and one-dimensional, the inversion is performed once per call by tabulating $\mathcal{F}$ on a uniform grid (default 4001 nodes over $[U_{10N}^{\mathrm{lo}}, U_{10N}^{\mathrm{hi}}]$) and interpolating linearly in the inverse direction. This agrees with a root-finding inversion to better than $10^{-5}\,\mathrm{m\,s^{-1}}$ and is orders of magnitude faster than a per-observation solve. If a supplied custom prior is not strictly monotone, its monotone envelope is used and a warning is issued.

Out-of-range friction velocities. Observed u_* outside $[\mathcal{F}(U_{10N}^{\mathrm{lo}}), \mathcal{F}(U_{10N}^{\mathrm{hi}})]$ has no preimage. Such rows are flagged in the output column `U10N_out_of_range` and, by default, clamped to the corresponding endpoint; optionally they are rejected instead. The flag is retained regardless of the row's subsequent fate, so the diagnostic count remains correct in both modes. The fraction

$$f_{\mathrm{oor}} = 100 \times \frac{N_{\text{out-of-range}}}{N_{\text{prior evaluated}}} \tag{21}$$

should be reported: if it is small the choice of prior range is a footnote, and if it is large that range becomes a scientific assumption of the analysis.

5.3 Charnock coefficient

The coefficient law is linear in the proxy wind, with saturation limits:

$$\alpha_{\mathrm{law}} = \mathrm{clip}\Big[a_1 \min\Big(\widehat{U}_{10N}, U_{\mathrm{cap}}\Big) + a_0,\ \alpha_{\min},\ \alpha_{\max}\Big]\,, \tag{22}$$

with defaults $a_1 = 0.0017$, $a_0 = -0.0050$, $\alpha_{\min} = 0.011$, $\alpha_{\max} = 0.028$, $U_{\mathrm{cap}} = 19\,\mathrm{m\,s^{-1}}$.

Below roughly $4\,\mathrm{m\,s^{-1}}$ the surface is aerodynamically smooth or transitional and a wind-dependent α is not well constrained by observation. Three low-wind treatments are offered:

$$\alpha = \begin{cases} \alpha_{\text{law}} & (\texttt{none}), \\ \alpha_{\min} \text{ if } \widehat{U}_{10N} < U_1, \text{ else } \alpha_{\text{law}} & (\texttt{hold}, \text{ default}), \\ \alpha_{\min} + S\!\left(\frac{\widehat{U}_{10N}-U_1}{U_2-U_1}\right)(\alpha_{\text{law}} - \alpha_{\min}) & (\texttt{ramp}), \end{cases} \tag{23}$$

where $S(x) = x^2(3-2x)$ is the smoothstep function on $[0,1]$ and $U_1 = 4$, $U_2 = 6\,\mathrm{m\,s^{-1}}$.

5.4 Momentum roughness closures

Five closures are available; exactly one is selected per call.

Option	Definition
`charnock_winddep`	$z_0 = \dfrac{\alpha(\widehat{U}_{10N})\,u_*^2}{g} + \dfrac{\beta_v \nu}{u_*}$ (default; Section 5.2)
`charnock_const`	$z_0 = \dfrac{\alpha_c\,u_*^2}{g} + \dfrac{\beta_v \nu}{u_*}$, α_c constant
`fixed`	$z_0 = z_0^{\text{fixed}}$, a prescribed constant
`eq3`	Drennan-type wave-age scaling, $z_0 = 3.35\,H_s \left(\dfrac{u_*}{c_p^w}\right)^{3.4}$
`yelland`	Taylor–Yelland wave-steepness scaling, $z_0 = 1200\,H_s \left(\dfrac{H_s}{L_p}\right)^{4.5}$ $[+\ \beta_v\nu/u_*]$

The wave closures require a sea state. Where the peak wavelength or phase speed is unavailable, deep-water relations may be used:

$$c_p^w = \frac{gT_p}{2\pi}, \qquad L_p = \frac{gT_p^2}{2\pi}. \tag{24}$$

These are valid for $h \gtrsim L_p/2$, i.e. a minimum depth $h_{\min} \approx 0.7804\,T_p^2$ (metres, T_p in seconds): about 112 m at $T_p = 12$ s and 176 m at $T_p = 15$ s. Applying them in shallower water is an error.

> **Warning.** The wave closures must be supplied with *per-observation* H_s and T_p. Driving them with a constant sea state makes z_0 nearly independent of u_* and inflates the apparent spread between roughness models at high wind by roughly a factor of four (Section 8).

5.5 Scalar roughness

Heat and moisture roughness lengths follow the COARE mapping from the roughness Reynolds number,

$$\mathrm{Rr} = \frac{z_0 u_*}{\nu}, \qquad z_{0t} = z_{0q} = \min\!\left(1.6\times10^{-4},\ \frac{5.8\times10^{-5}}{\mathrm{Rr}^{0.72}}\right). \tag{25}$$

Two structural properties of Eq. (25) are important.

Rr **has a hard floor.** On the Charnock branches, substituting z_0 gives the identity

$$\mathrm{Rr} = \beta_v + \frac{\alpha\,u_*^3}{g\,\nu}, \tag{26}$$

so $\mathrm{Rr} \geq \beta_v = 0.11$ always. The aerodynamically smooth limit is therefore $\mathrm{Rr} \to 0.11$ *from above*, and a criterion of the form $\mathrm{Rr} < 0.11$ is unsatisfiable by construction. A practical near-smooth threshold is $\mathrm{Rr} < 0.135$, at which the Charnock term contributes under 20%. Fully rough flow, $\mathrm{Rr} > 2$, begins at $u_* \approx 0.292\,\mathrm{m\,s^{-1}}$, i.e. $U_{10N} \approx 9\,\mathrm{m\,s^{-1}}$.

The cap severs the dependence on z_0. The first branch of Eq. (25) binds for $\mathrm{Rr} < (5.8 \times 10^{-5}/1.6 \times 10^{-4})^{1/0.72} = 0.2443$, equivalently $u_* < 0.121\,\mathrm{m\,s^{-1}}$ or $U_{10N} \lesssim 4.0\,\mathrm{m\,s^{-1}}$. In that regime z_{0t} is a constant and, since z_0 influences the fluxes *only* through z_{0t}, the heat fluxes become exactly independent of the momentum roughness. Every physical roughness closure returns the same flux there.

> **Remark.** Three independently motivated thresholds coincide near $U_{10N} \approx 4\,\mathrm{m\,s^{-1}}$: the low-wind Charnock cutoff $U_1 = 4.0$; the crossover at which the viscous term ceases to dominate z_0, at 4.3; and the scalar-roughness cap boundary, at 4.01. The low-wind treatment is therefore sitting on a physical transition rather than on a tuning constant.

6 Numerical solution

6.1 Structure

Because u_* is prescribed, z_0, z_{0t}, z_{0q} and Rr are all determined *before* the iteration begins and are loop-invariant. Only (θ_*, q_*, ζ) are iterated. The fixed-point map is

$$\mathbf{x} = (\theta_*, q_*), \qquad \mathbf{x}^{(k+1)} = \mathcal{G}\Big(\mathbf{x}^{(k)}\Big), \tag{27}$$

where $\mathcal{G}$ evaluates ζ from Eq. (5) and then (θ_*, q_*) from Eqs. (2)–(3). The initial guess is the neutral solution $\psi_h \equiv 0$.

6.2 Convergence criterion

The residual is measured on the *raw* fixed-point step, before any relaxation is applied, and combines a relative and an absolute floor:

$$d = \max\left(\frac{|\mathcal{G}_\theta(\mathbf{x}) - \theta_*|}{|\theta_*| + \epsilon_a},\ \frac{|\mathcal{G}_q(\mathbf{x}) - q_*|}{|q_*| + \epsilon_a}\right) < \mathrm{tol}, \tag{28}$$

with defaults $\mathrm{tol} = 10^{-6}$, $\epsilon_a = 10^{-8}$. Measuring d on the raw step matters: if it were computed from the relaxed update $\mathbf{x}^{(k+1)} - \mathbf{x}^{(k)} = \lambda\,(\mathcal{G} - \mathbf{x})$ the residual would scale with λ, making the convergence threshold depend on the relaxation coefficient and rendering the reported residual incomparable between passes.

6.3 Row freezing

Rows are solved simultaneously but independently: once a row satisfies Eq. (28) it is frozen and no longer updated, even while other rows continue to iterate. Without freezing, a row that converged at iteration 8 would continue to be advanced until the slowest row in the batch finished, so its result would depend on which other observations happened to share the call. Freezing guarantees that batched and individually solved results are bit-identical, which is a reproducibility requirement when a record is processed in chunks.

6.4 Under-relaxation

Plain Picard iteration fails for strongly stable rows at low u_*. The algorithm therefore performs a second pass, on the failed rows only, with under-relaxation

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \lambda \left[\mathcal{G}\left(\mathbf{x}^{(k)}\right) - \mathbf{x}^{(k)} \right], \qquad \lambda = 0.5. \tag{29}$$

Rows that already converged are untouched, so their values are unchanged. In a representative stable case ($T_s = 10°\mathrm{C}$, $T_a = 22°\mathrm{C}$) this raises the convergence rate from 4/6 to 6/6, with the damped rows converging in 8–20 iterations.

6.5 The free-convection limit

Under-relaxation does not rescue every row, and should not be expected to. Since $\zeta \propto u_*^{-2}$, the stability parameter diverges as $u_* \to 0$:

u_* ($\mathrm{m\,s^{-1}}$)	$\lvert\zeta\rvert$	iterations	outcome
0.004	1.5×10^4	–	never converges (λ from 1.0 to 0.02)
0.010	553	30	converges
0.030	36	25	converges
0.060	7.4	23	converges
0.130	1.3	21	converges

This is the free-convection limit, in which Monin–Obukhov similarity at fixed u_* is ill-posed rather than merely difficult. The correct response is rejection, not a more aggressive solver. Setting the admissible minimum $u_*^{\min} \approx 0.06\,\mathrm{m\,s^{-1}}$ ($U_{10N} \approx 2\,\mathrm{m\,s^{-1}}$) removes such rows in advance.

7 Validity and quality control

7.1 Admissibility

A row enters the solver only if $u_* > u_*^{\min}$ (strictly; a vanishing friction velocity has no physical surface-layer solution and would yield a roughness generated solely by numerical floors), all state variables are finite, and the measurement heights exceed the corresponding roughness lengths. Inadmissible rows are returned as `NA` rather than aborting the call.

7.2 A non-converged solve is not a flux

If a row fails to satisfy Eq. (28) within the iteration limit, **every solver-dependent quantity is blanked**, and the logical column `flux_valid` is set `FALSE`. The output is partitioned as follows.

Class	**Columns**
Blanked on failure	`HS`, `HL`, `L`, `zeta_zu`, `Ug`, `gf`, `Umean_zu`
Retained as debug state	`tstar`, `qstar`, `n_iter`, `final_residual`, `used_relaxation` — these describe the *unconverged* state and must not be analysed as results
Retained, solver-independent	`tau`, `z0`, `zot`, `zoq`, `Rr`, `alpha_used`, `U10N_proxy`, `U10N_out_of_range`, `U10N_diag`, `rho_a`, `Q`, `Qs`, `dt`, `dq`

Note that $\tau = \rho_a u_*^2$ and $U_{10N}^{\mathrm{diag}} = (u_*/\kappa)\ln(10/z_0)$ never touch the iteration and so remain valid. Downstream analysis should filter on `flux_valid`.

7.3 Reported diagnostics

The quality-control summary distinguishes three population counts that are easily conflated:

$$N_{\text{flux valid}} = N_{\text{solver}} - N_{\text{non-converged}}, \tag{30}$$

where N_{solver} counts rows that reached the iteration. Reporting only N_{solver} would allow a failed solve to be counted as a usable observation. Alongside these, the out-of-range fraction f_{oor}, the number of rows requiring under-relaxation, the maximum final residual, and the flow-regime counts are reported.

8 Sensitivity and uncertainty budget

8.1 How the roughness reaches the flux

With u_* prescribed, z_0 does not appear in Eq. (1), nor in the stability parameter Eq. (5). Its *only* path to the flux is

$$z_0 \;\longrightarrow\; \mathrm{Rr} = \frac{z_0 u_*}{\nu} \;\longrightarrow\; z_{0t} \;\longrightarrow\; \ln(z_t/z_{0t}) \;\longrightarrow\; \theta_*. \tag{31}$$

Because the dependence is logarithmic, the elasticity is small:

$$\frac{\partial \ln H_S}{\partial \ln z_{0t}} = \frac{1}{\ln(z_t/z_{0t})} \approx 0.07\text{–}0.13, \qquad \frac{\partial \ln H_S}{\partial \ln z_0} \approx -0.72 \times \frac{\partial \ln H_S}{\partial \ln z_{0t}}. \tag{32}$$

A factor-of-ten error in z_0 therefore costs only 10–16% in H_S, and where the cap of Eq. (25) binds it costs nothing at all.

8.2 Elasticities

Computed on a 2×10^4-observation synthetic record (Weibull wind field, $T_s - T_a = 3\,\mathrm{K}$), the elasticities $\partial \ln H_S / \partial \ln x$ are:

U_{10N} bin	n	$\Delta\theta$	u_*	z_{0t}	z_0
0–4	2825	1.095	0.790	0.1323	0.0000
4–6	3876	1.077	0.659	0.1142	−0.0821
6–8	4209	1.059	0.683	0.1002	−0.0721
8–10	3598	1.041	0.712	0.0893	−0.0643
10–12	2360	1.027	0.721	0.0803	−0.0578
12–15	1782	1.017	0.757	0.0737	−0.0531
15–20	585	1.010	0.788	0.0683	−0.0492
>20	34	1.005	0.850	0.0641	−0.0462

The air–sea temperature difference has unit elasticity: a 10% error in $\Delta\theta$ is a 10% error in H_S, at every wind speed. The friction velocity has elasticity 0.66–0.85; it is sub-linear because raising u_* also raises z_0, which lowers z_{0t} and partially offsets the direct effect. The momentum roughness has elasticity ≈ 0.05 — roughly **twenty times less important** than $\Delta\theta$ per unit fractional error.

In the lowest bin $\partial \ln H_S / \partial \ln z_0$ is exactly zero while $\partial \ln H_S / \partial \ln z_{0t} = 0.13$. These are consistent: the cap makes z_{0t} unresponsive to z_0, but the flux still depends on z_{0t} itself. The practical implication is that for that portion of the record the low-wind heat flux is controlled entirely by the constant 1.6×10^{-4} in Eq. (25), and by no roughness model at all.

8.3 Roughness-model ensemble spread

Running the four physical closures over the same record and taking the spread per observation gives the systematic uncertainty attributable to the choice of roughness model:

U_{10N} bin	n	$\overline{H_S}$ (W m^{-2})	spread (%)	median z_0 ratio
0–4	2826	15.73	**0.00**	36.4×
4–6	3876	22.68	3.43	11.5×
6–8	4209	28.69	9.28	6.3×
8–10	3598	35.78	**10.90**	4.9×
10–12	2360	44.54	8.74	4.2×
12–15	1782	55.27	7.39	3.8×
15–20	585	70.09	6.49	3.6×
>20	34	90.07	5.89	3.5×

Three features deserve comment. First, the spread is *exactly* zero in the lowest bin despite a 36× disagreement in z_0 itself — the cap of Section 5.5 absorbs it entirely. Second, the spread peaks in mid-range winds and *declines* at high wind, provided the wave closures are driven with a wind-consistent sea state; with a constant sea state the high-wind spread is inflated to over 20%, which is an artefact of the forcing rather than a property of the closures. Third, the latent and sensible spreads are equal to within rounding, because $z_{0t} = z_{0q}$ in Eq. (25); consequently the **Bowen ratio is invariant** to the choice of roughness model even where the individual fluxes differ by 11%.

8.4 Guidance

Accuracy is best purchased in the following order:

1. $\Delta\theta$, i.e. the SST and air-temperature fields. Unit elasticity at all winds; a 0.5 K bias on a 3 K gradient is comparable to the entire roughness-model spread.
2. The retrieved u_* itself, at elasticity ≈ 0.7–0.85.
3. Per-observation H_s and T_p, which convert the wave closures from assumptions into constrained estimates and cut the high-wind ensemble spread by roughly a factor of four.
4. The scalar roughness z_{0t} — not z_0 — ideally constrained against direct flux measurements. Fixing z_{0t} to within a factor of two caps the induced flux error at about 7%.

The roughness-model systematic should be reported *stratified by wind*, not as a single number.

9 Usage reference

9.1 Invocation

```
source("bulk_algo_no_circularity_with_prior_v2.R")

out <- flx_compute_HS_HL_fixed_ustar_prior(
  u_star = ustar,            # prescribed friction velocity [m/s]
  T_s    = sst,              # sea surface temperature [degC]
  T_a    = tair,             # air temperature [degC]
  RH     = rh,               # relative humidity [%]
  P      = pres,             # pressure [hPa or Pa; auto-detected]
  zt = 10, zq = 10, zu = 10, # heights [m]; scalar or length-n
  z0_model   = "charnock_winddep",
```

```r
  prior_model = "andreas",
  u_star_min = 0.06          # reject the free-convection tail
)

flux <- subset(out, flux_valid)   # ALWAYS filter on this
flx_qc_summary(out)               # QC roll-up for a paper table
```

9.2 Principal arguments

Argument	Default	Meaning
`u_star`	—	prescribed friction velocity, length n
`T_s, T_a, RH, P`	—	bulk state; scalar or length n
`zt, zq, zu`	10	measurement heights; scalar or length n
`S, lat, zi`	35, 45, 600	salinity, latitude, inversion height
`z0_model`	`charnock_winddep`	one of `charnock_winddep`, `charnock_const`, `fixed`, `eq3`, `yelland`
`prior_model`	`andreas`	`andreas`, `edson2013` or `custom`
`prior_custom_fun`	`NULL`	user function $U_{10N} \mapsto u_*$
`Hs, Tp, Lp, cp`	`NA`	sea state for the wave closures
`u_star_min`	0	admissible lower bound on u_*
`clamp_inversion`	`TRUE`	clamp (vs. reject) out-of-range u_*
`alpha_lowwind_mode`	`hold`	`hold`, `ramp`, `none`
`scalar_rough_fn`	`NULL`	override for Eq. (25); signature `function(rr, visa, u_star)` returning `list(zot=, zoq=)`
`cp_moist`	`FALSE`	use $c_{pv} = c_p(1 + 0.84Q)$
`use_gustiness`	`TRUE`	compute U_g, g_f (diagnostic only)
`maxit, tol`	60, 10^{-6}	iteration controls
`relax_retry`	0.5	damping for the second pass; `NULL` disables

9.3 Principal outputs

Column	Meaning
`HS, HL`	sensible and latent heat flux, $\mathrm{W\,m^{-2}}$, upward positive
`flux_valid`	**use this filter**; `FALSE` where the solve failed
`tau`	$\rho_a u_*^2$; a restatement of the input
`tstar, qstar`	scaling parameters (unconverged state if `flux_valid` is `FALSE`)
`L, zeta_zu`	Obukhov length and z_u/L
`z0, zot, zoq, Rr`	roughness lengths and roughness Reynolds number
`alpha_used`	Charnock coefficient actually applied
`U10N_proxy`	$\mathcal{F}^{-1}(u_*)$; a relabelling of the input
`U10N_out_of_range`	u_* outside the prior's representable range
`U10N_diag, Umean_zu`	diagnostic winds; **never** regress against these
`Ug, gf`	gustiness diagnostics; cannot affect `HS`/`HL`
`converged, n_iter`	convergence flag; iterations performed
`final_residual`	$\|\mathcal{G}(\mathbf{x}) - \mathbf{x}\|$ at exit
`used_relaxation`	whether the damped second pass was applied
`rho_a, Q, Qs, dt, dq`	thermodynamic intermediates

Scalar metadata (roughness model, prior, height ranges, and the quality-control roll-up) is attached as attributes: `attr(out, "qc")`, `attr(out, "z0_model")`, and so on.

10 Summary of assumptions

1. u_* is an observation and is not adjusted. All momentum-related outputs are diagnostic.
2. The prior $\mathcal{F}$ is independent of the present computation. If it were calibrated on the same data the anti-circularity guarantee of Section 5.1 would be void.
3. $\widehat{U}_{10N}$ is a monotone relabelling of u_*, not an independent wind estimate, and carries no additional information.
4. Deep water is assumed wherever deep-water wave relations are used.
5. Surface-layer Monin–Obukhov similarity is assumed to hold. It does not in the free-convection limit (Section 6.5); such rows are rejected.
6. No Webb correction, cool-skin, warm-layer, or rain heat flux.

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